

MCP Calling - Overview - Z.AI DEVELOPER DOCUMENT

Source: <https://docs.z.ai/guides/llm/glm-5-turbo>

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MCP Calling

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MCP server calling is based on the Model Context Protocol (MCP) standard, dynamically connecting and calling tools and resources provided by external MCP servers in chat/completions. Through MCP server calling, you can extend the model's capability boundaries, implementing various functions such as search, visual understanding, file processing, data analysis, and more, providing powerful tool ecosystem support for AI applications.

Features

The MCP server calling feature integrates MCP tool calling capabilities in chat, supporting direct calls to chat/completions to connect and use various MCP servers to extend model functionality without requiring an MCP client. Through MCP server calling, AI can:

Dynamic Tool Discovery: Automatically discover and enumerate tools provided by MCP servers

Real-time Tool Calling: Call various tools and functions from external MCP servers

Multi-protocol Support: Support SSE and Streamable-HTTP transport protocols

Flexible Configuration: Support custom server addresses, authentication information, and tool permissions

Zhipu Ecosystem: Built-in official Zhipu MCP servers, direct use without URL configuration

Standard Compatibility: Fully compatible with MCP protocol standards, supporting third-party MCP servers

Supported Transport Protocols

SSE (Server-Sent Events)

Use Cases: Real-time data streams, long connection scenarios

Features: Unidirectional data flow, server-initiated push

Simple Configuration: Pass authentication information through URL parameters

Streamable-HTTP

Use Cases: Standard HTTP request-response

Features: Bidirectional communication, supports complex interactions

Flexible Configuration: Support custom Headers and parameters

Parameter Description

server_label

stringrequired

MCP server identifier. If connecting to official Zhipu MCP servers, use mcp code to fill this field, no need to fill server_url

server_url

string

MCP server address. Required when connecting to third-party MCP servers

transport_type

stringdefault:"streamable-http"

Transport protocol type

sse: Server-Sent Events protocol

streamable-http: Standard HTTP protocol (default)

allowed_tools

array

Collection of allowed tools to call. If not specified, all available tools are allowed
["webSearchPrime", "image_analysis", "video_analysis"]

headers

object

Authentication information and custom headers required by MCP server

```
{
  "Authorization": "Bearer your_api_key",
  "Custom-Header": "custom_value"
}
```

Code Examples

cURL

Python

Connect to Official Zhipu MCP Server

```
curl --location 'https://api.z.ai/api/paas/v4/chat/completions' \
--header 'Authorization: Bearer YOUR_API_KEY' \
--header 'Content-Type: application/json' \
--data '{
  "model": "glm-4.6",
  "messages": [
    {
      "role": "user",
      "content": "Search for the latest artificial intelligence development trends"
    }
  ],
  "tools": [
    {
```

```

        "type": "mcp",
        "mcp": {
            "server_label": "mcp code",
            "transport_type": "streamable-http",
            "allowed_tools": ["webSearchPrime"]
        }
    },
    "tool_choice": "auto"
}'

```

Connect to Third-party MCP Server

```

curl --location 'https://api.z.ai/api/paas/v4/chat/completions' \
--header 'Authorization: Bearer YOUR_API_KEY' \
--header 'Content-Type: application/json' \
--data '{
    "model": "glm-4.6",
    "messages": [
        {
            "role": "user",
            "content": "Analyze the content of this document"
        }
    ],
    "tools": [
        {
            "type": "mcp",
            "mcp": {
                "server_label": "document-analyzer",
                "server_url": "https://doc-mcp.example.com/mcp",
                "transport_type": "streamable-http",
                "headers": {
                    "Authorization": "Bearer doc_analysis_token",
                    "X-Custom-Header": "custom_value"
                },
                "allowed_tools": ["analyze_document", "extract_text"]
            }
        }
    ],
    "tool_choice": "auto"
}'

```

Using SSE Protocol

```

curl --location 'https://api.z.ai/api/paas/v4/chat/completions' \
--header 'Authorization: Bearer YOUR_API_KEY' \
--header 'Content-Type: application/json' \
--data '{
    "model": "glm-4.6",
    "messages": [
        {
            "role": "user",
            "content": "Get real-time weather information"
        }
    ],
    "tools": [
        {

```

```
    "type": "mcp",
    "mcp": {
      "server_label": "weather-service",
      "server_url": "https://weather-mcp.example.com/sse?token=weather_token",
      "transport_type": "sse",
      "allowed_tools": ["get_current_weather", "get_forecast"]
    }
  },
  "tool_choice": "auto"
}'
```

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